

THE HONG KONG POLYTECHNIC UNIVERSITY
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Tutorial 1

Vibe Coding Essentials

COMP5511 AI Conceptions

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Beginning?

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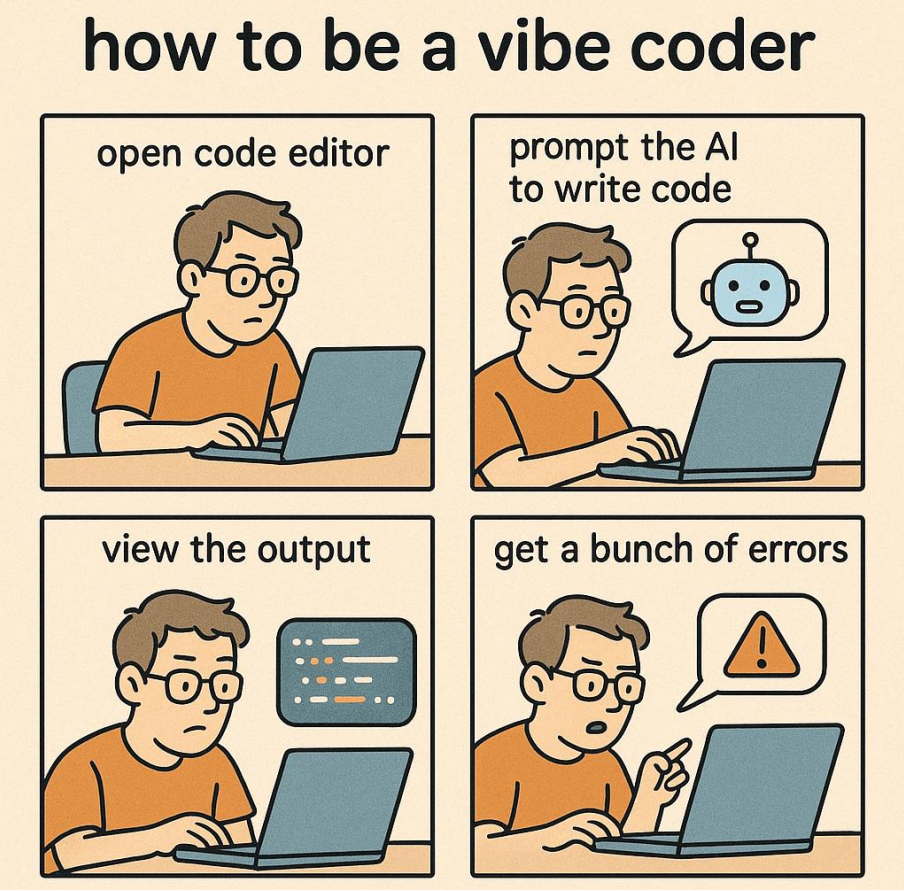
~~"Talk is cheap, show me the code"~~

"forget that the code even exists"

Andrej Karpathy, February 2025

Natural-language requests drive code generation and iteration.
The original description accepts limited code inspection.

What is vibe coding?



The promise

Describe a task. Get a first result.

The next step

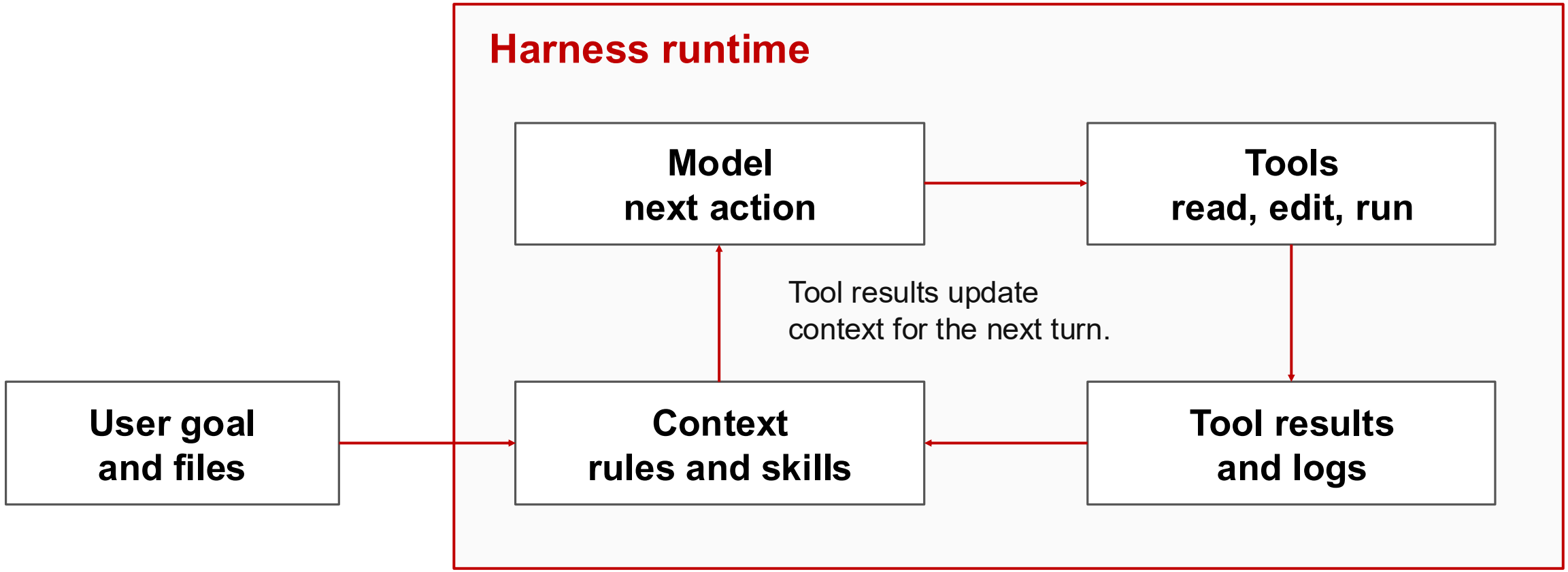
(Let LLM) Inspect what happened when it fails.

What is vibe coding?



Dimension	Traditional	AI-assisted (e.g. GPT)	Vibe coding (e.g Codex)
Main input	Code	Code and prompts	Natural language
AI role	No	Suggest or complete	Generate and iterate
Human effort	Design and implement	Implement and review	Specify and inspect results
Main failure mode	Implementation bugs Don’t know how to code	Unchecked suggestions	Hidden errors accumulate

The agent harness

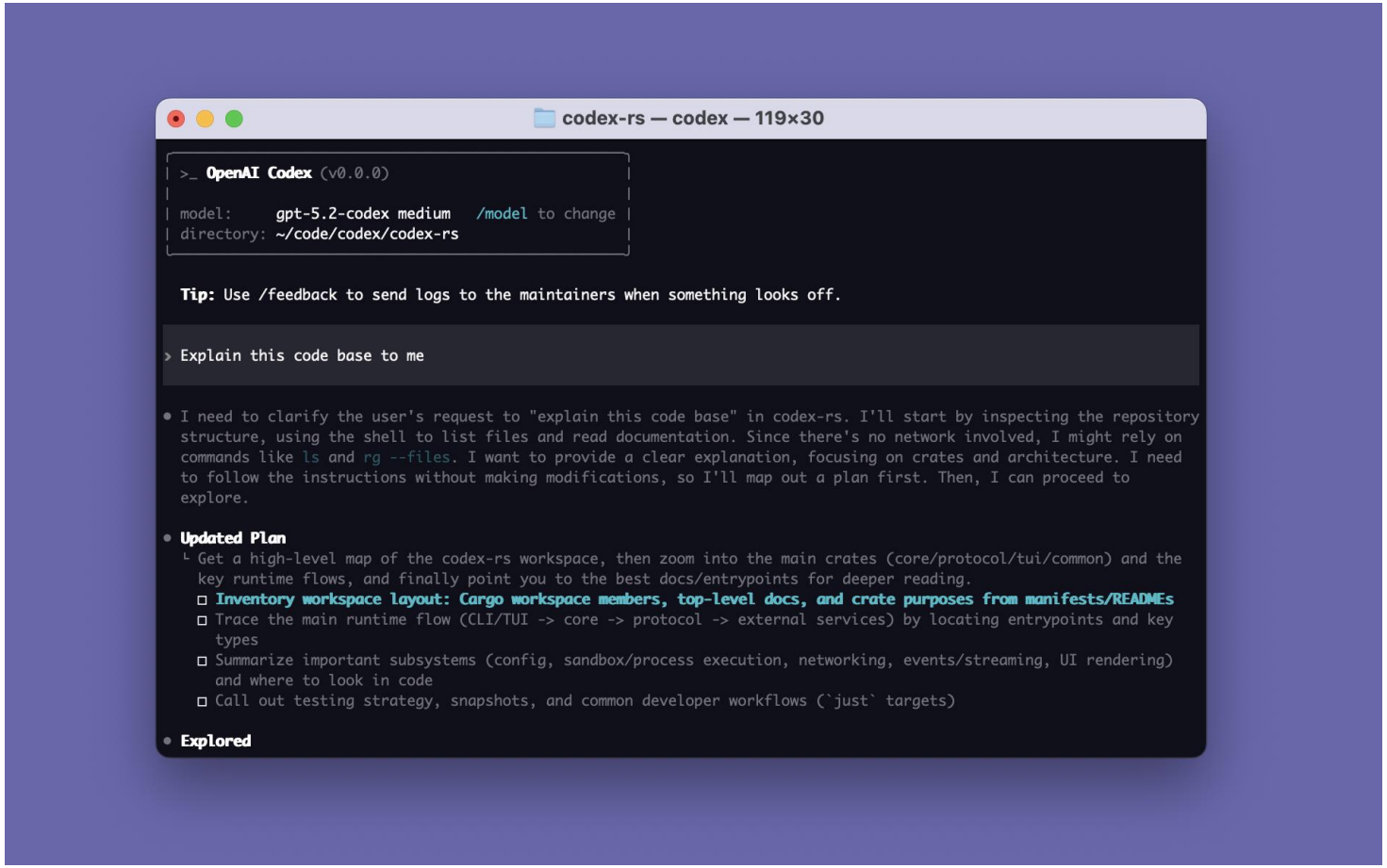


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AI coding tools

A quick visual tour before we use WorkBuddy

Codex



Where you work

App, CLI or IDE

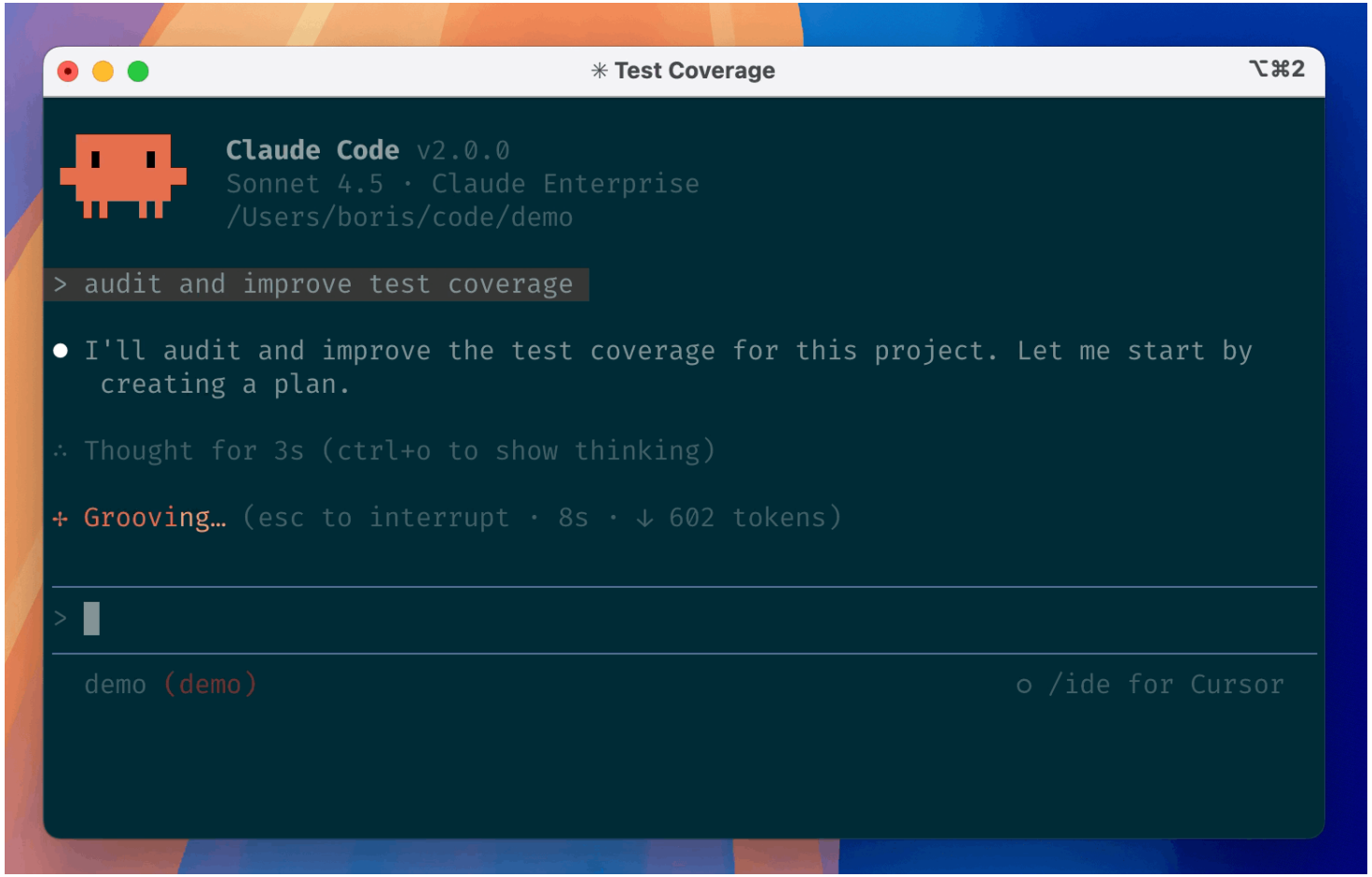
Agent workflow

Reads project files, edits code and runs commands.

Example task

Fix a plotting script and run it.

Claude Code



Where you work

Terminal and IDE

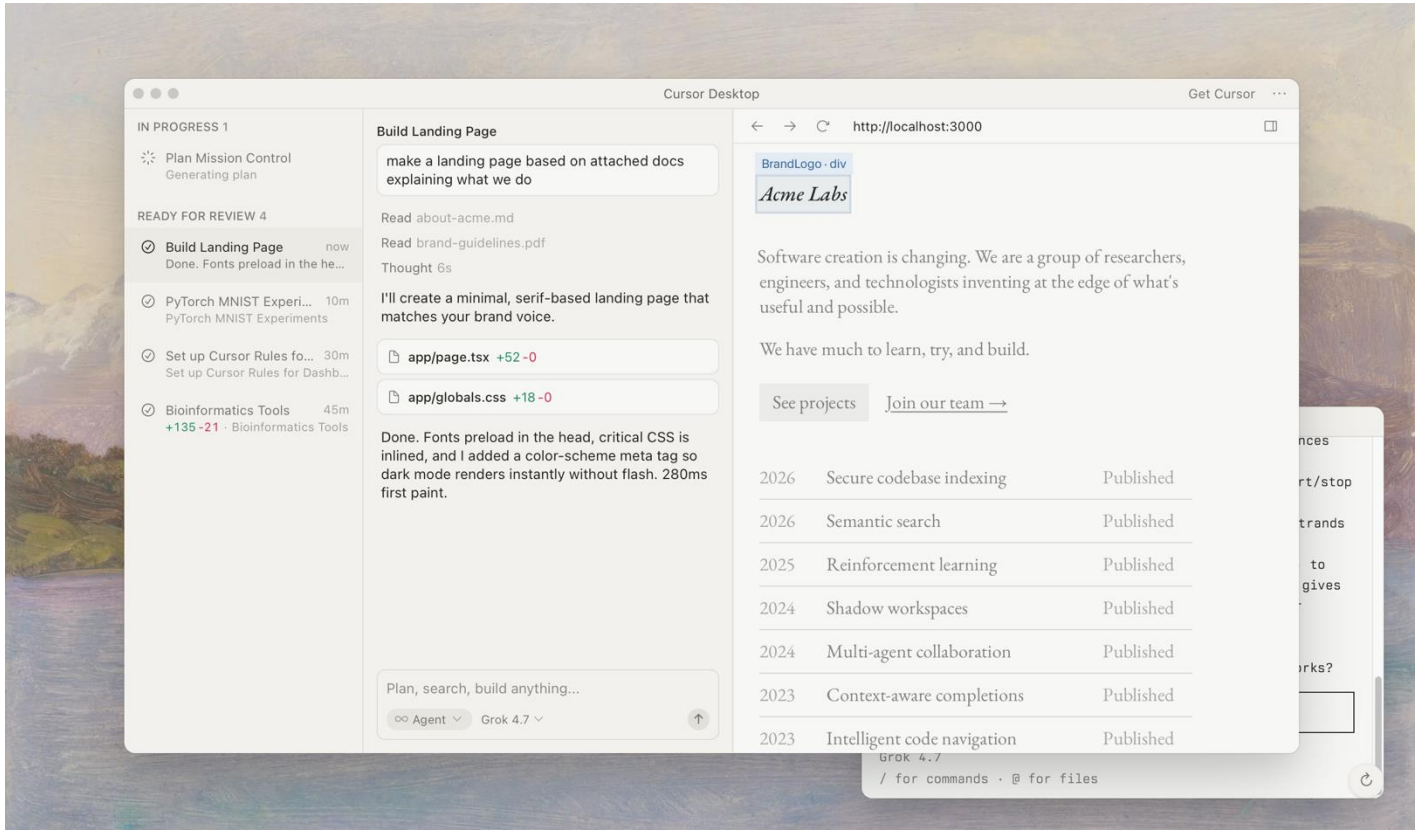
Agent workflow

Explores a codebase and uses tools to carry out a coding task.

Example task

Audit a project's test coverage.

Cursor



Where you work

Desktop or Web UI

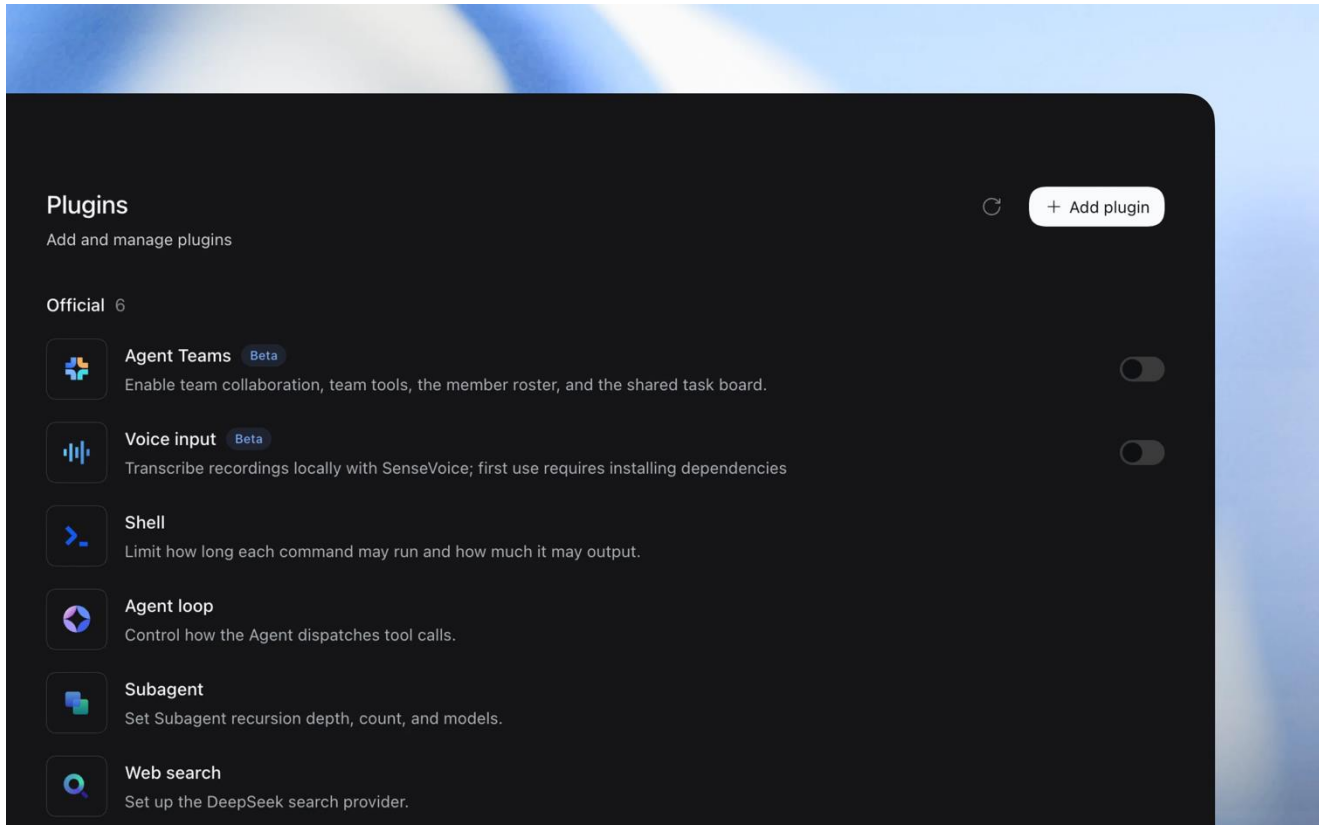
Agent in the editor

The agent searches the codebase, edits files and runs terminal commands.

What you check

The diff and the working preview.

DeepSeek Harness



Where you work

Desktop or Web UI

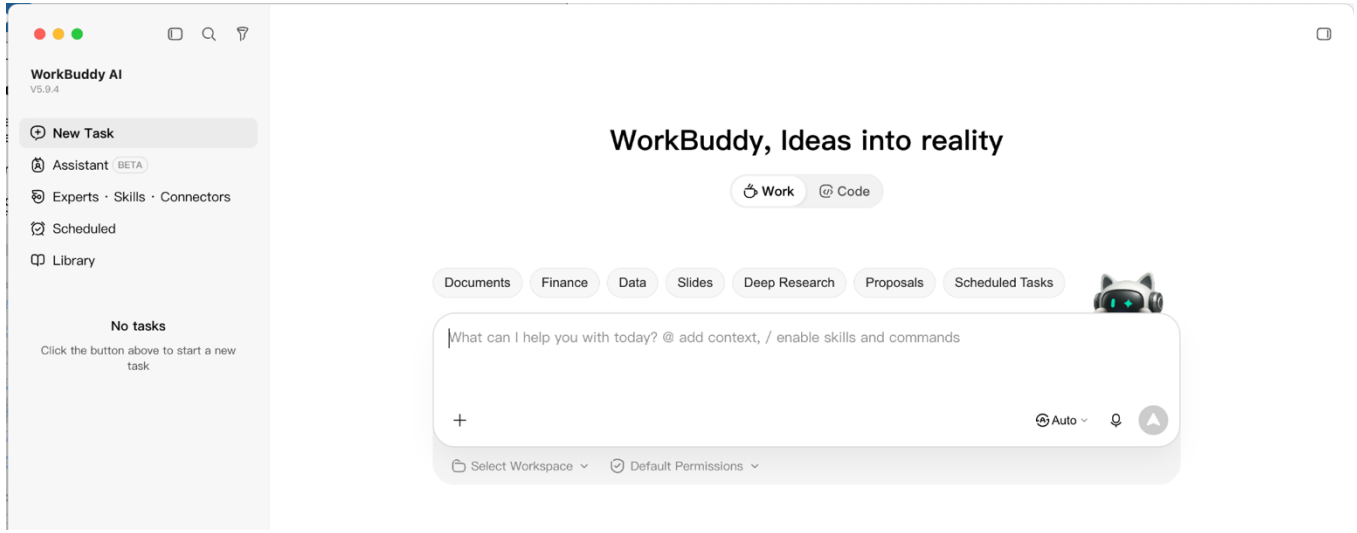
Harness design

An open source harness with composable plugins.

What you inspect

Available plugins and the agent's tool trace.

WorkBuddy



A workspace for agent tasks

The selected folder gives the agent access to the project context.

Built-in skill entry

Find, upload or create a skill from the Skills interface.

What you check

The actual output files and preview.

Coding tools at a glance

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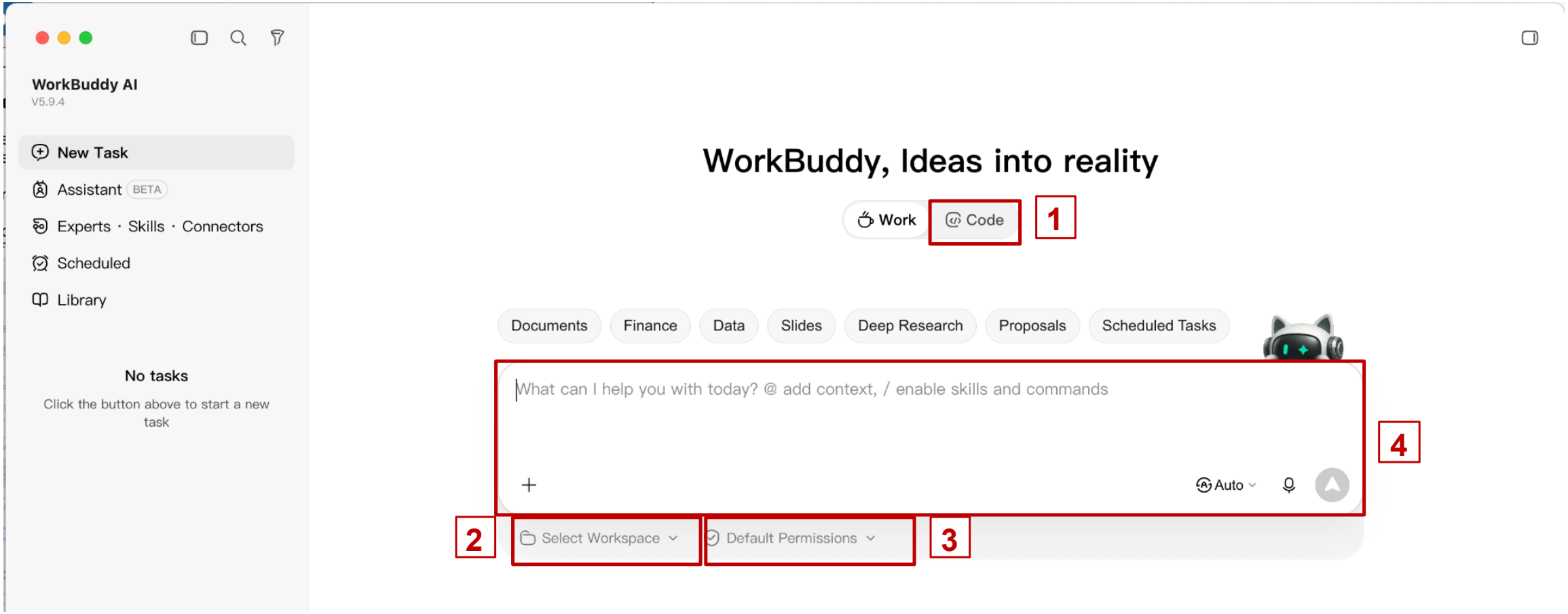
Tool	Typical surface
Codex	App, CLI and IDE
Claude Code	Terminal, IDE and desktop
Cursor	Editor with an agent
DeepSeek Harness	Desktop or Web UI
WorkBuddy	Desktop workstation

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WorkBuddy

Workspace, permissions, models and files

The WorkBuddy interface



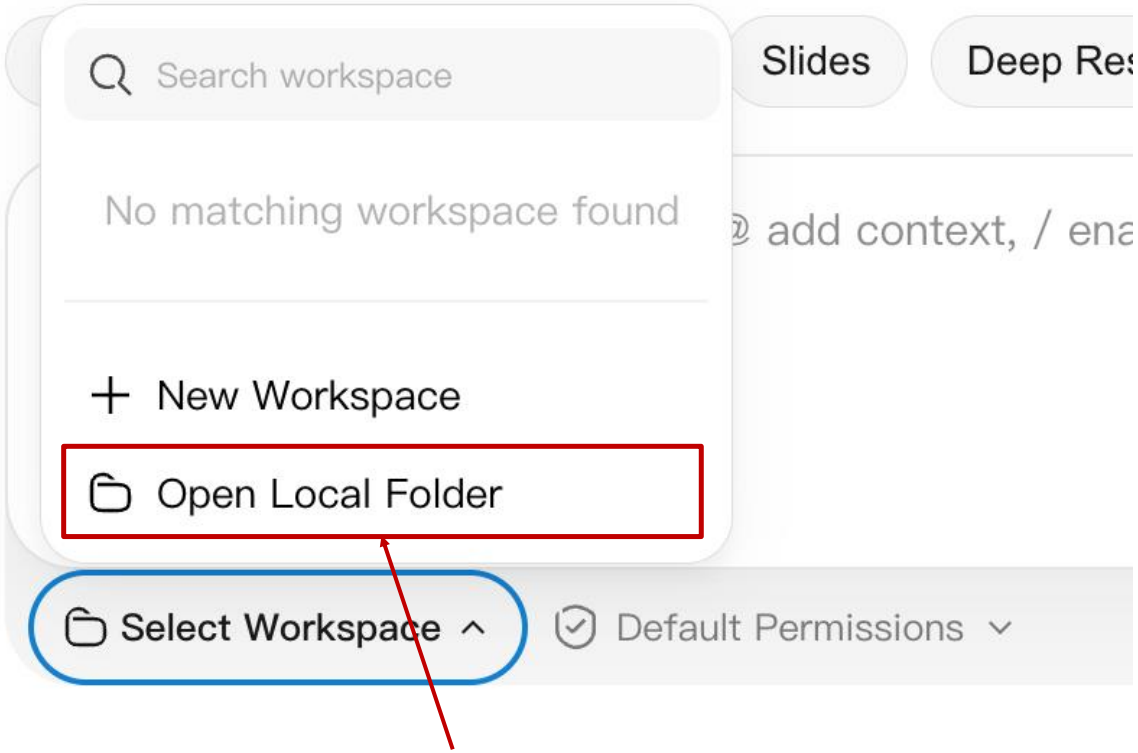
1 Code mode

2 Workspace

3 Permissions

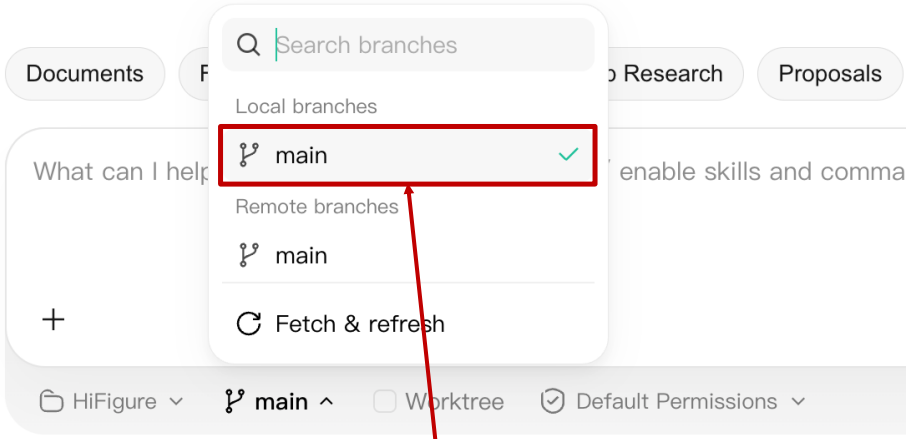
4 Prompt input

Workspace and Git branch



1 Open the course folder

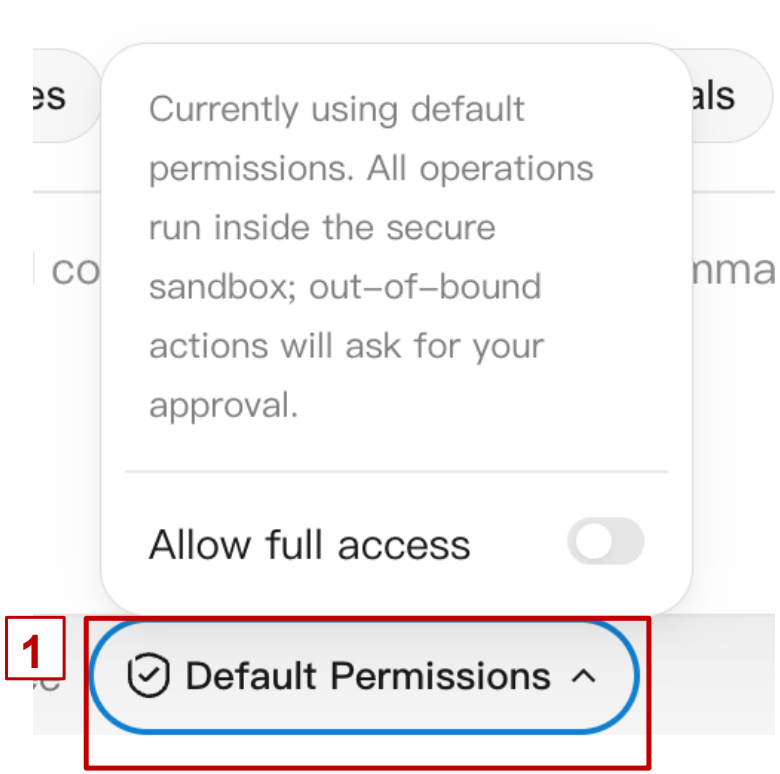
The workspace folder supplies the file context for the task.



2 Current Git branch

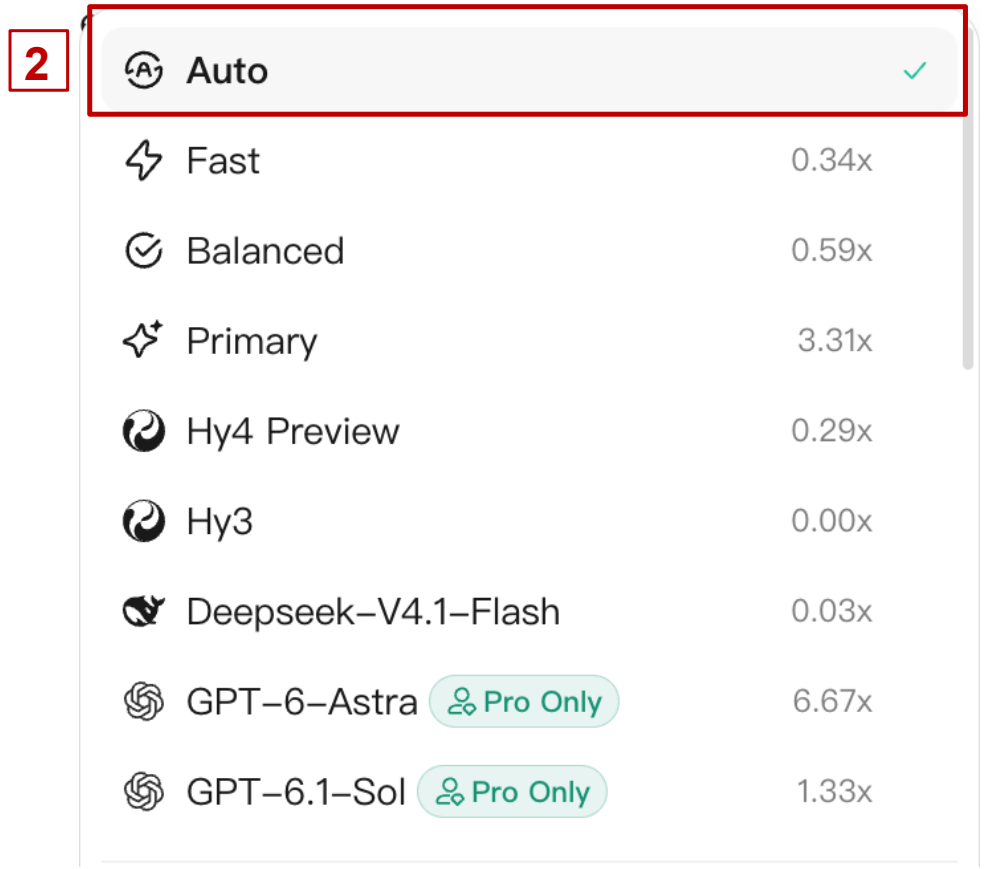
Git controls appear for a repository.

Permissions and model choice



Default Permissions

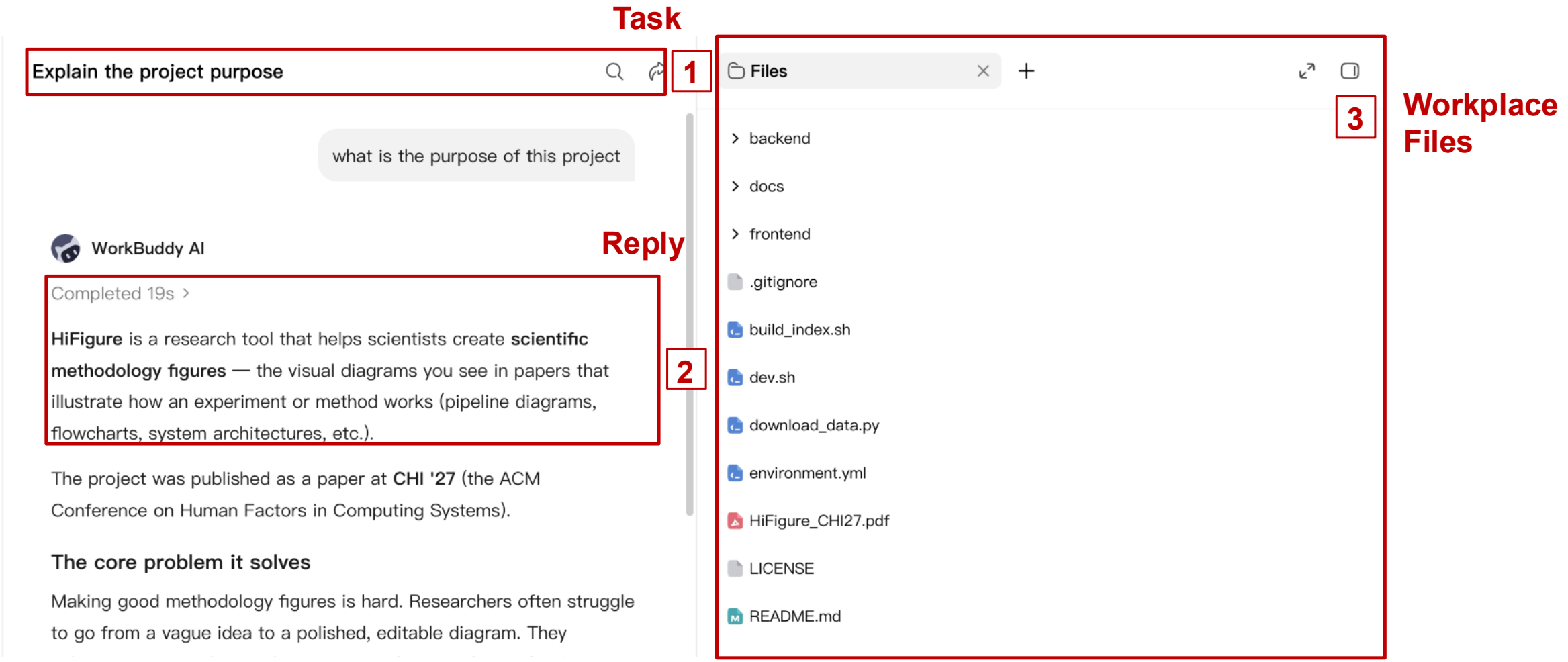
Full access stays off for this exercise.



Available models

Options and usage multipliers can change.

Conversation and project files



Exercise 0

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Download Materials from Github via Vibe Coding

3 minutes

Input https://github.com/Pangjing-Wu/COMP5511

01 Open a workspace folder.

02 Ask WorkBuddy to Download it.

03 Check the download files.

Exercise 1

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Inspect a workspace

4 minutes

Input toy_results.csv

01 Open the course folder as your workspace and locate toy_results.csv.

02 Ask WorkBuddy to report the CSV column names and row count without editing files.

03 Open the CSV yourself and compare it with the agent's answer.

Produce A reply identifying the real columns and number of rows.

Check The answer agrees with the file you opened.

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Prompting

A clear request and a visible success condition

Five prompt elements

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Goal	What should the program do?
Context	Which files, columns and environment should it use?
Constraints	Which methods and limits apply?
Deliverables	Which files should the agent produce?
Acceptance	How will you verify that it worked?

Two prompts for the same task

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algorithm	run	step	score
Algorithm A	0	0	0.4536566049570532
Algorithm A	0	10	0.6853221568723644
Algorithm A	0	20	0.8181519812520827
Algorithm A	0	30	0.8704636976164589
Algorithm A	0	40	0.8582445859458893
Algorithm A	0	50	0.8761318084177218
Algorithm A	0	60	0.8978306986659484
Algorithm A	0	70	0.8945410502194597
Algorithm A	0	80	0.8990506853370203
Algorithm A	0	90	0.8894275092553475
Algorithm A	0	100	0.9104018175157977
Algorithm A	1	0	0.4593335032251474
Algorithm A	1	10	0.6985943345179849

Prompt A

Make a beautiful chart from this CSV.

What is missing?

Which columns should be used?
Which files should it save?
How can we verify the result?

Prompt B

Goal
Compare score by training step.

Context
Read toy_results.csv and inspect its columns.

Constraints
Use matplotlib and only observed values.

Deliverables
Save plot.py, and figure.pdf only.

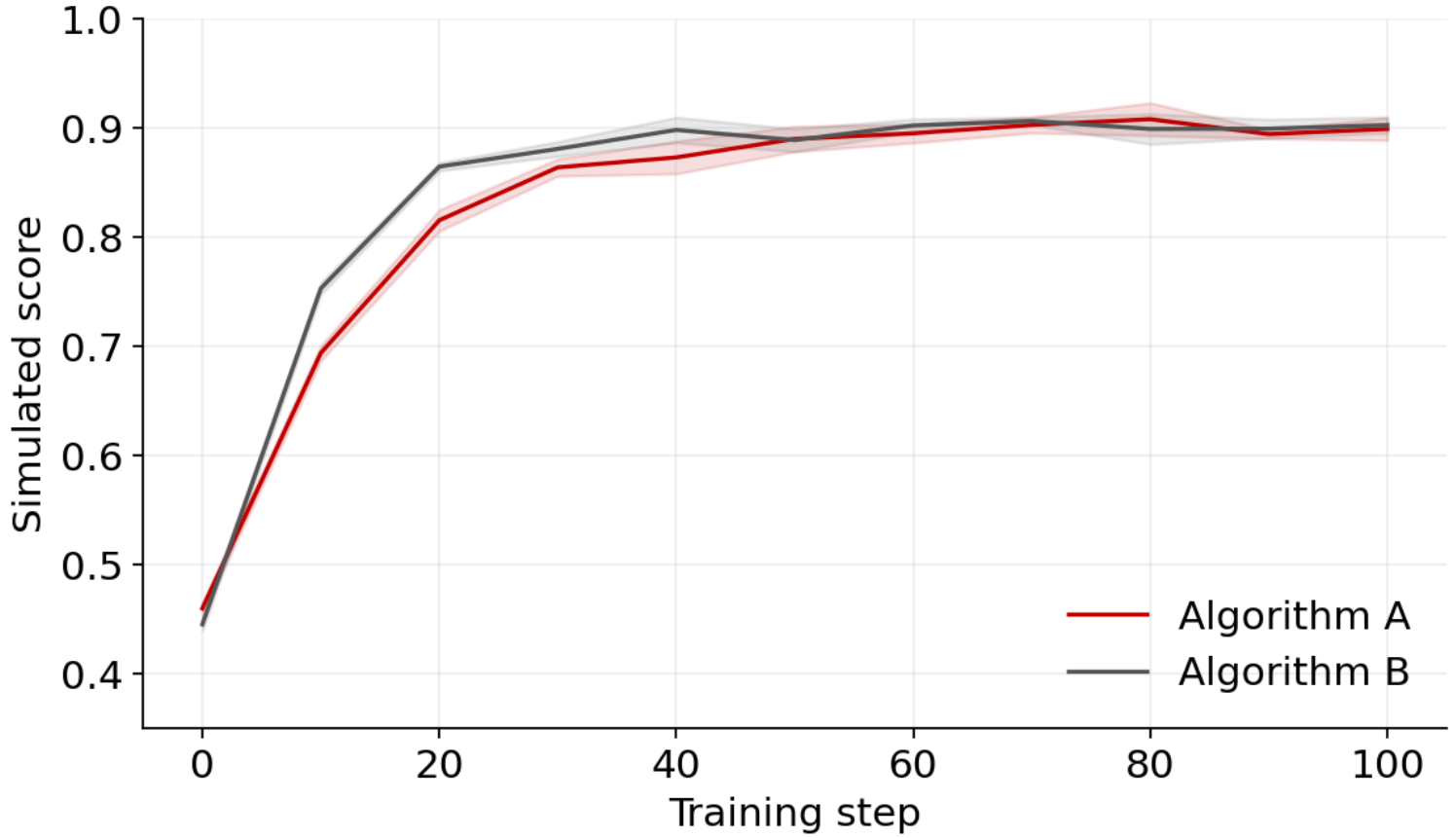
Acceptance
Run the script and inspect the saved figure.

A structured figure prompt

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Goal	Compare algorithms by training step.
Context	Read toy_results.csv. Inspect the columns and missing values first.
Constraints	Use pandas and matplotlib. Show mean ±1 SD only for repeated runs. Use observed values only.
Deliverables	Save plot.py, and figure.pdf only.
Acceptance	Run plot.py. Open the figure and check the axis labels and error-band meaning.

Lab data and target output



Input columns

algorithm, run, step, score

Data size

2 algorithms and 11 steps per run

Interpretation

These values do not support a real algorithm comparison.

Exercise 2

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Write and run your own prompt

8 minutes

Input toy_results.csv and the five-element prompt guide

01 Write your own prompt. Include all five elements and specify the actual input file.

02 Run the task in WorkBuddy. Open the generated script and figure.

03 Compare prompts with a partner and identify one requirement you would improve.

Produce Your prompt, plot.py and a saved figure.

Check The script re-runs and the figure uses the observed CSV values.

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Skills

Reusable instructions for recurring tasks

What is a skill?

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Reusable procedure

A skill stores instructions for a recurring kind of task.

When to use it

Its name and description help the agent decide when it is relevant.

Optional resources

A skill may include scripts, references and templates.

Task-specific input

You still provide the file, the goal and the success condition.

A small research skill

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```
---
name: research-figure
description: Plot CSV experiment results reproducibly.
---

Inspect column names, types and missing values.
Use only measurements present in the file.
Label axes and state any aggregation.
Show spread only when repeated runs exist.
Save the script, PNG and SVG.
Run the script and report errors.
```

SKILL.md

Metadata describes when to use the skill.

Instructions

The body defines the procedure.

Optional resources

Scripts, references and example data can accompany it.

Finding useful skills

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Official framework guidance

Streamlit official skills

https://docs.streamlit.io/develop/api-reference/cli/skills

Official examples

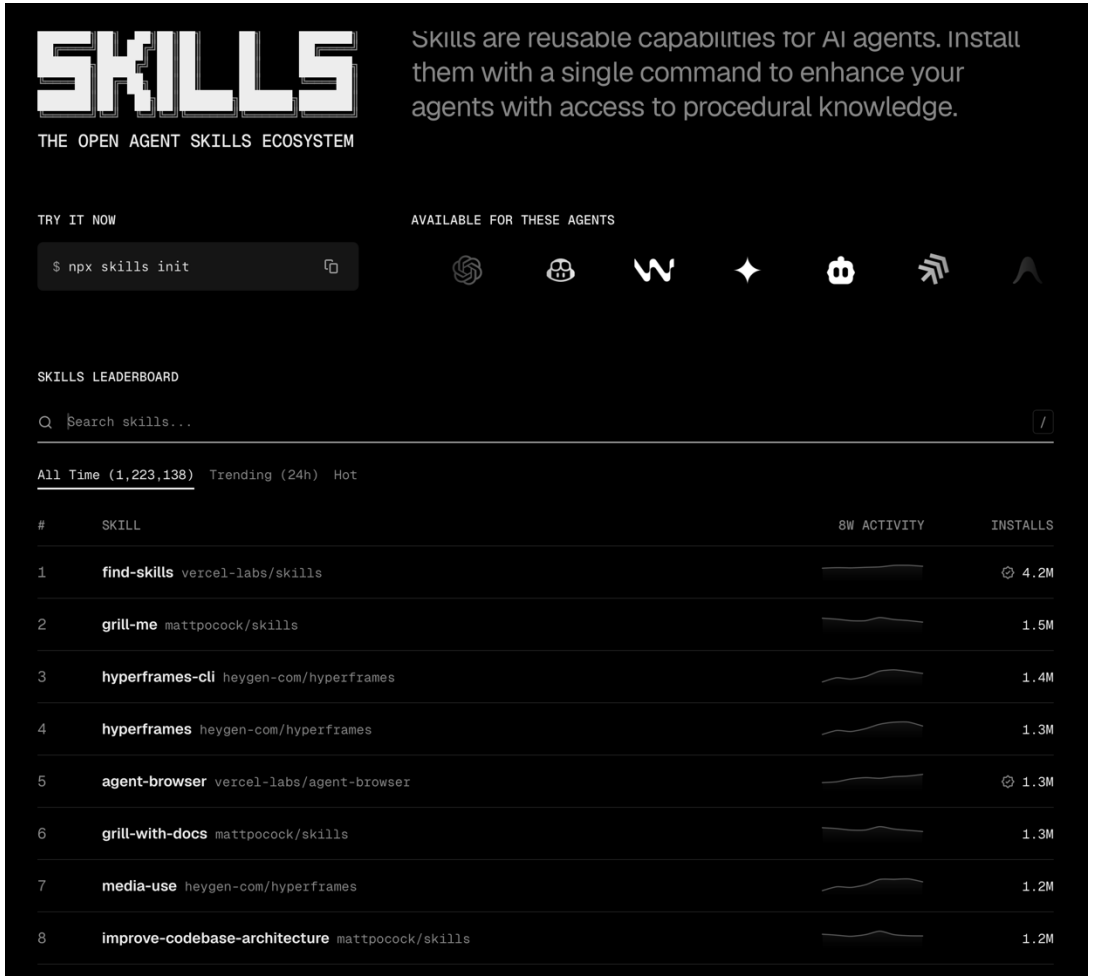
OpenAI Skills and Anthropic Skills

https://github.com/openai/skills https://github.com/anthropics/skills

Community discovery

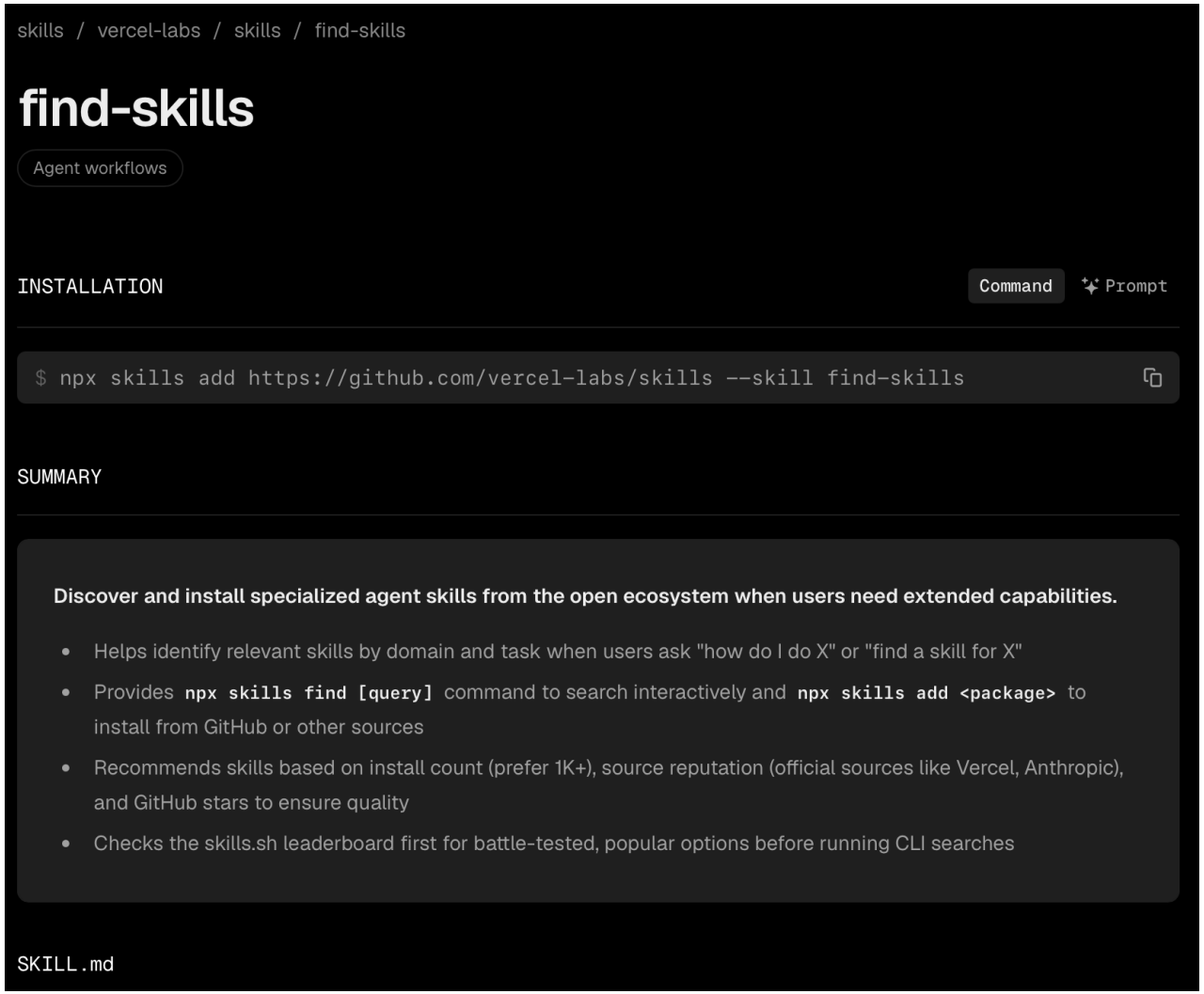
skills.sh. Follow the link to the source repository.

https://skills.sh/



Finding useful skills

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Use the skill to find skills

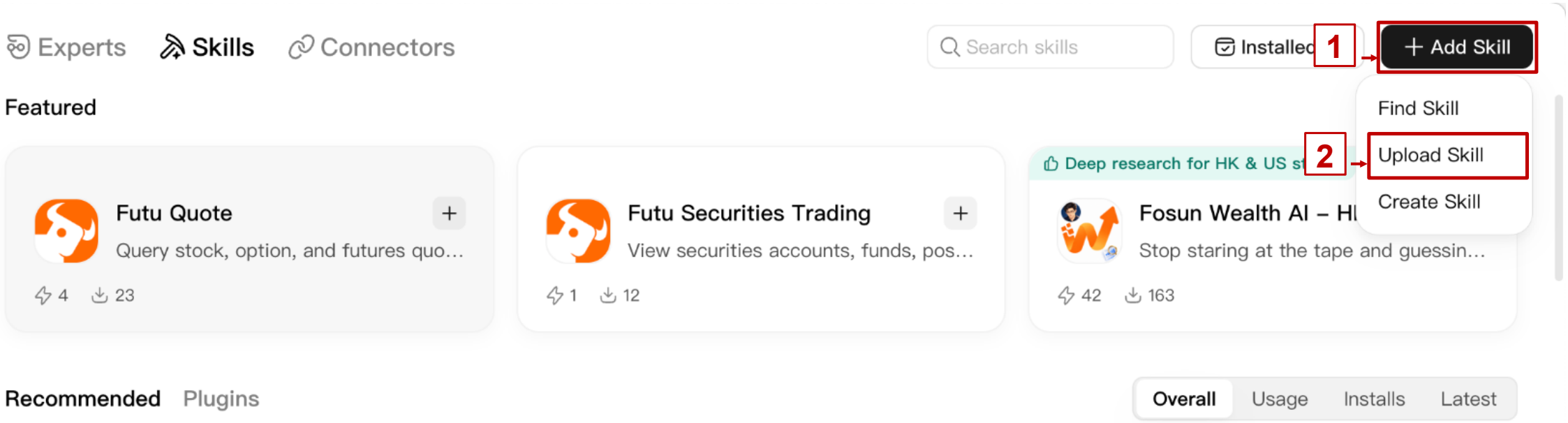
Example response:

```
I found a skill that might help! The "react-best-practices" skill provides
React and Next.js performance optimization guidelines from Vercel Engineering.
(185K installs)

To install it:
npx skills add vercel-labs/agent-skills@react-best-practices

Learn more: https://skills.sh/vercel-labs/agent-skills/react-best-practices
```

Installing a skill in WorkBuddy



Find Skill

Search for a workflow.

Upload Skill

Import a local skill package.

Create Skill

Describe a reusable procedure.

This exercise uses Add Skill, then Upload Skill.

A prompt with a reusable skill

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Repeated prompt

Inspect the columns and missing values.
Use the observed data only.
Compare algorithms by training step.
Label the axes and aggregation.
Show spread only for repeated runs.
Save a script, PNG and SVG.
Run the script and report errors.

With research-figure

Use research-figure skill to plot toy_results.csv.
Save outputs in skill_output/.

The installed skill supplies the reusable plotting procedure.

Exercise 3

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Install the research-figure skill

3 minutes

Input research-figure.zip

01 Open Skills and choose Add Skill, then Upload Skill.

02 Import research-figure.zip and read its name, description and instructions.

03 Find research-figure in your installed skills and confirm that it is enabled.

Produce An installed and enabled research-figure skill.

Check You can locate the installed skill and explain when to use it.

Exercise 4

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Use the skill with a short prompt

3 minutes

Input The installed research-figure skill and toy_results.csv

01 Select or invoke research-figure in a task using the same workspace.

02 Use a short request. Ask it to plot toy_results.csv and save outputs in skill_output/.

03 Open the output and check whether the reusable instructions were followed.

Produce A figure and script generated using the skill.

Check The short request still produces a runnable script and a clearly labelled figure.

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Debugging

A small fix and a reproducible check

A useful debugging request

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Observed failure

State the exact command and paste the complete error message.

Expected behavior

Explain what should have happened and which input file exists.

Bounded change

Ask for the smallest fix that addresses the cause.

Verification

Run the same command again and inspect the changed lines.

For the next exercise, run plot_bug.py and diagnose its input path.

Exercise 5

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Fix an error and verify the change

6 minutes

Input plot_bug.py and toy_results.csv

01 Run plot_bug.py and give WorkBuddy the exact command and error message.

02 Ask for a minimal fix, then inspect the changed lines.

03 Re-run the script and check the output labels and error band.

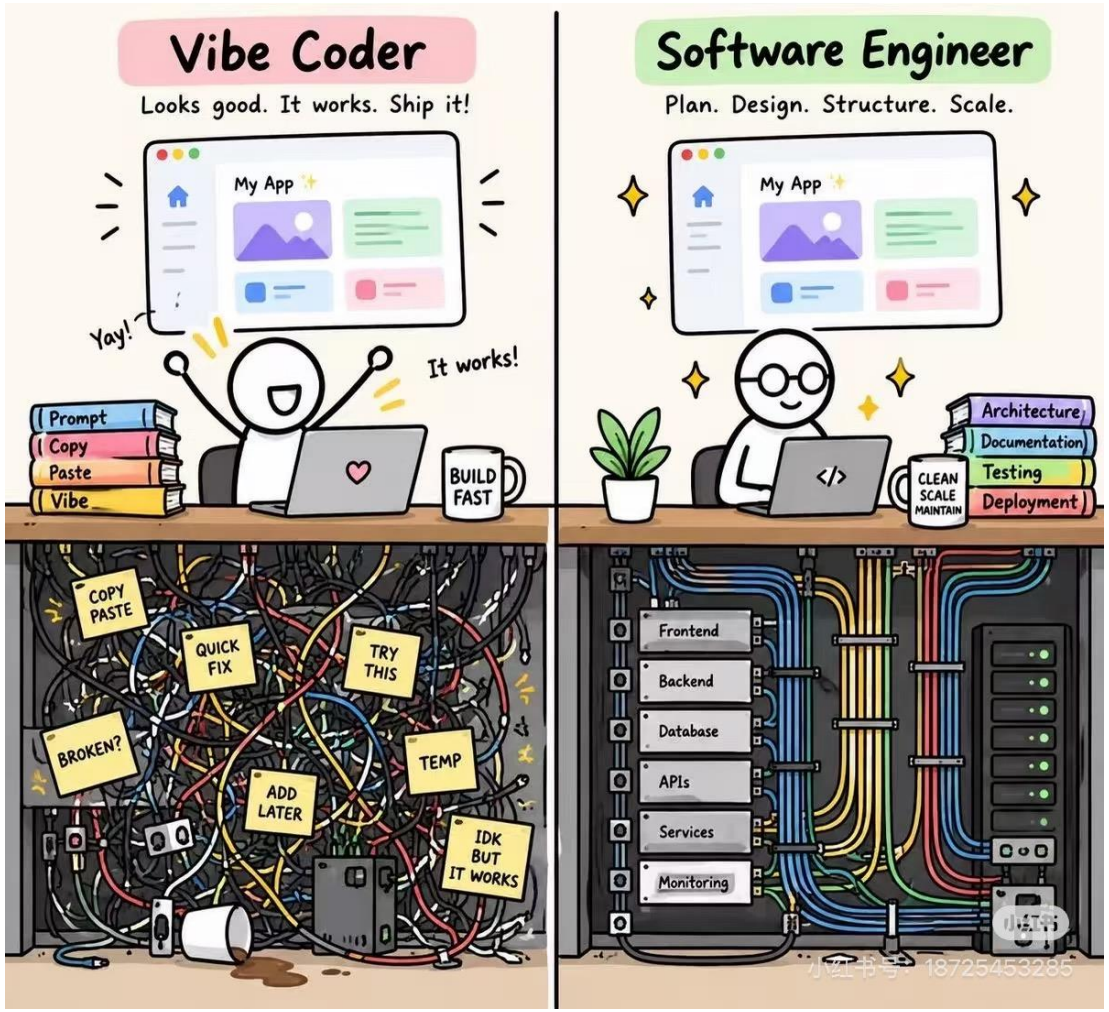
Produce The corrected script, a figure and a short explanation of the error.

Check The original failing command succeeds after a change you can explain.

Bitter Lesson

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Do code review, know what you want and what the agents did.





Further reading in Vibe Coding

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Hook

Event trigger

Run a validation script after code edits.

Configure PostToolUse to run python plot.py.
Save the hook in .codex/hooks.json.

main codex-plugin-cc / plugins / codex / hooks / **hooks.json**

dkundel-openai and **codex** Initial commit

Code Blame 38 lines (38 loc) · 845 Bytes

```
{
  "description": "Optional stop-time review gate for Codex Companion.",
  "hooks": {
    "SessionStart": [
      {
        "hooks": [
          {
            "type": "command",
            "command": "node \"${CLAUDE_PLUGIN_ROOT}/scripts/session-lifecycle-hook.mjs\" SessionStart",
            "timeout": 5
          }
        ]
      }
    ],
```

Further reading in Vibe Coding

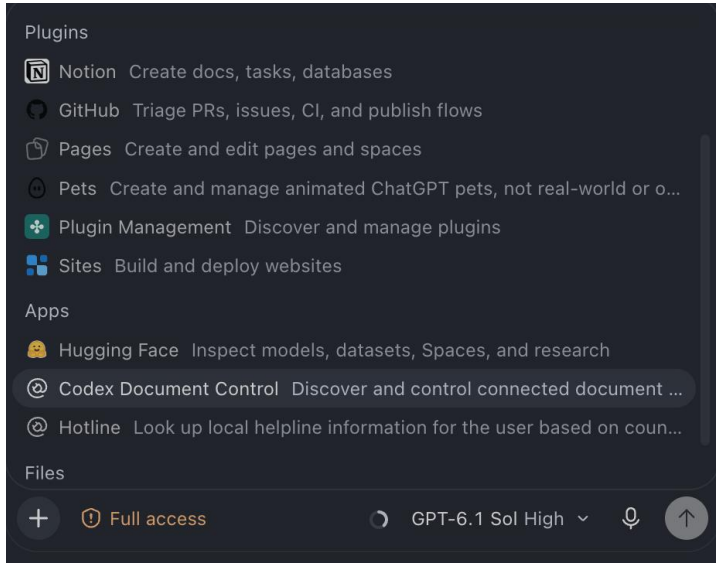
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MCP

External tools

Connect Codex to official documentation.

```
codex mcp add docs --url
https://developers.openai.com/mcp
```



Subagent

Parallel work

Delegate independent reviews to separate agents.

“Spawn two subagents. One reviews plot.py, the other checks toy_results.csv. Wait for both and summarize their findings.”

Tutorial 1 recap

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Workspace

Which folder and files did the agent actually read?

Prompt

Which of the five elements changed your result?

Skill

Which instructions can you now reuse?

Verification

What evidence shows that the script works?